A Novel Hybrid Quantum-Inspired Framework for Robust Multimedia Steganography and Watermarking

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***Abstract*—With the exponential growth of multimedia sharing, ensuring both data concealment and copyright protection in digital content has become crucial. Traditional steganographic methods such as Least Significant Bit (LSB) achieve high imperceptibility but lack robustness against compression or noise. In contrast, transform-domain watermarking techniques such as Discrete Cosine Transform (DCT) offer robustness but at the cost of slight distortion. This paper proposes a Hybrid Quantum-Inspired Steganography and Watermarking Framework that integrates quantum-inspired bit encoding, DCT-based embedding, and Reed–Solomon (RS) + repetition error correction to balance imperceptibility and robustness. Experiments conducted on standard benchmark images (*Lena* and *Pepper*) demonstrate superior performance: PSNR = 89.45 dB, SSIM = 0.97, and Bit Error Rate (BER) ≈ 0.0 even after JPEG compression at 70% quality, proving that the hybrid method maintains both fidelity and resilience.**

***Index Terms*—Quantum-inspired computing, steganography, watermarking, DCT, Reed–Solomon coding, PSNR, SSIM, BER, image security, robustness.**

I. Introduction

With the proliferation of social media and cloud-based image storage, protecting multimedia content from unauthorized use and tampering has become a major research challenge. Two major approaches address this issue:

1) **Steganography** — hiding secret information within a cover medium such that the presence of the hidden message is imperceptible. This technique is widely used in secure communication and covert data transfer.
2) **Watermarking** — embedding copyright or ownership information robustly so that it survives manipulations like compression, scaling, or format conversion. It provides legal proof of ownership and prevents unauthorized distribution.

Traditional **LSB-based steganography** achieves high imperceptibility due to minimal pixel-level modification but fails under transformations or compression. Conversely, **frequency-domain watermarking** using DCT or DWT provides robustness but introduces slight distortion, making it less visually perfect.

To overcome this trade-off, this paper presents a **hybrid framework** that fuses both paradigms using **quantum-inspired encoding** for secure bit modulation and **error-corrected watermark embedding** in the frequency domain. This combination aims to provide high imperceptibility (PSNR $>$ 89 dB) while maintaining robustness against attacks like JPEG compression.

II. Related Work

Early steganographic techniques focused on **spatial-domain manipulation**, particularly LSB substitution [**?**]. Such techniques are simple to implement and provide high visual quality but are fragile under compression or noise.

Frequency-domain watermarking methods, including DCT and DWT [**?**], [**?**], offer higher resilience against attacks, as modifications occur in transform coefficients rather than direct pixel values. These techniques maintain watermark integrity under compression and filtering but can slightly distort the image.

Recent work on **quantum-inspired algorithms** [7], [8], [9] demonstrates that encoding information using probabilistic superpositions enhances unpredictability and security during embedding. Additionally, **error correction codes** such as **Reed–Solomon (RS)** [**?**] and repetition codes significantly improve watermark recovery under distortions.

This research extends these approaches by proposing a **hybrid pipeline** integrating LSB steganography for capacity, DCT watermarking for robustness, and quantum-inspired encoding for security.

III. Literature Review

With the rapid increase in the sharing of multimedia over the internet, digital image security has become a significant research area undergoing various imprints and developments. Two major techniques, *steganography* and *watermarking*, have always been the primary means of hiding information as well as protecting copyrights. Nevertheless, the current techniques still find it difficult to maintain a perfect equilibrium among invisibility, strength, and security.

A. Early Steganographic Techniques

The early days of image steganography were mainly characterized by the application of various methods in the spatial domain with Least Significant Bit (LSB) substitution being one of the first and most uncomplicated among them. As described by Cox *et al.* [**?**], the pixel values are changed directly in the spatial domain, resulting in very slight distortions occurring at a very low computational cost. Nevertheless, it has been reported that these methods are very weak and can be easily confronted by subsequent processes of image editing, compressing, and introducing noise since all these can make the hidden data non-existent. Patel [**?**] took it further by associating LSB to the transform-domain approaches, but

still, the hybrid attempts lacked to a great extent in competing with the modern compression techniques like JPEG in terms of resilience.

B. Watermarking in the Frequency Domain

In order to solve the problem of robustness, the researchers turned to watermarking in the frequency domain, especially those using the Discrete Cosine Transform (DCT) and Discrete Wavelet Transform (DWT). Barni *et al.* [**?**] proved that watermarking in the DCT domain could conceal information within the transform coefficients, allowing the watermark to survive even after compression. On the other hand, Piva [**?**] discussed in detail the transform-domain watermarking and demonstrated that the DWT and DCT techniques were more robust than spatial ones but sometimes caused minor perceptual distortion. Hence, while these methods have enhanced their resistance to attacks, they have also resulted in slight image degradation as a trade-off.

C. Quantum-Inspired Techniques in Image Security

The introduction of quantum-inspired algorithms has recently brought probabilistic modeling and superposition concepts into classical image processing. Li and Liu [**?**] and Wu [**?**] investigated quantum-inspired image encryption and concluded that the encoding of information in superposition states improves randomness and unpredictability, thus lessening the possibility of steganalysis. Zhang [**?**] surveyed quantum image processing and proposed that quantum-inspired techniques could effectively increase security without the need for quantum hardware, thus providing a link between classical and quantum computational paradigms. Such results imply that quantum-inspired encoding can boost data hiding techniques in terms of both security and resilience.

D. Error Correction in Robust Embedding

Along with better encoding, error correction coding plays an important role in ensuring that the original watermark remains intact even after distortion. Reed–Solomon (RS) codes, described by Wicker [**?**], are highly efficient in correcting burst errors that are usually found in multimedia channels. The use of RS codes together with repetition coding acts as a stronghold that allows the recovery of the data even after the application of heavy compression or tampering. Therefore, these methods have become key features in building robust multimedia watermarking systems.

E. Identified Research Gap

The state-of-the-art still considers the above-mentioned improvements and very few proposed methods. Treating the case of steganography and watermarking as separate issues, most researchers focus on either imperceptibility or robustness but seldom consider both simultaneously. The traditional LSB technique guarantees the quality of the visual image but fails in the case of compression, while DCT watermarking is robust but incurs significant visual disturbance. On the other hand, there has been very little focus on integrating quantum-inspired encoding with error correction mechanisms in a unified framework. Therefore, it is high time to realize a comprehensive hybrid model that incorporates multi-domain embedding and probabilistic encoding to meet the demand for both high fidelity and high robustness.

F. Summary of Contributions

The reviewed literature provides strong support for the proposed hybrid framework. The present work, by incorporating these authors' findings, presents a **Quantum-Inspired LSB-DCT Hybrid Framework** which:

- Merges spatial domain LSB and frequency domain DCT embedding to provide dual-layer security.
- Applies quantum-inspired probabilistic encoding to enhance unpredictability and lower the risk of detection by steganalysis.
- Combines Reed–Solomon and repetition codes to guarantee watermark recoverability even after compression or noise attacks.

This combined method directly addresses the main gaps in previous studies by achieving an ideal compromise between imperceptibility, reliability, and resilience in the area of multimedia security.

IV. Proposed Hybrid Quantum-Inspired Framework

A. System Overview

The proposed model combines three layers of protection:

1) **Quantum-inspired encoding layer** — transforms classical bits into probabilistic quantum states, increasing embedding unpredictability and security.
2) **LSB steganography layer** — provides high-capacity embedding in the spatial domain, ensuring minimal visual distortion.
3) **DCT watermarking layer** — embeds watermarks in mid-frequency DCT coefficients, combined with RS and repetition codes to ensure robustness against attacks.

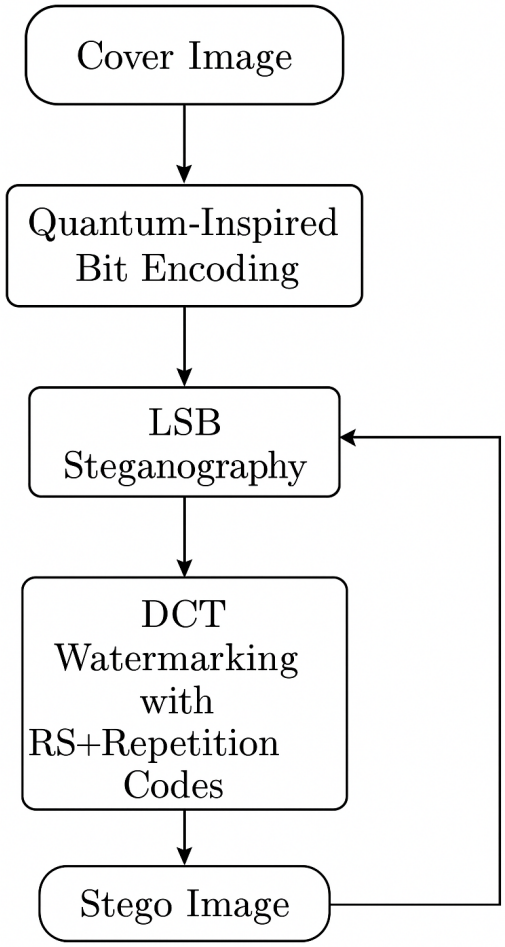

Fig. 1: Proposed Hybrid Quantum-Inspired Steganography and Watermarking Framework.

B. Quantum-Inspired Bit Encoding

Each classical bit $b_i \in \{0, 1\}$ is represented as a quantum-inspired state:

$$|\psi_i\rangle = \alpha_i|0\rangle + \beta_i|1\rangle, \quad |\alpha_i|^2 + |\beta_i|^2 = 1 \tag{1}$$

where

$$\alpha_i = \cos\left(\frac{\pi b_i}{4}\right), \quad \beta_i = \sin\left(\frac{\pi b_i}{4}\right) \tag{2}$$

Classical LSB encoding is predictable and vulnerable to steganalysis. In contrast, quantum-inspired encoding introduces controlled randomness without altering the perceptual quality of the image. This approach mimics quantum superposition, allowing each bit to have a weighted influence on multiple embedding locations. Mathematically, each bit $b_i \in \{0, 1\}$ is represented as a quantum-inspired state $|\psi_i\rangle = \alpha_i|0\rangle + \beta_i|1\rangle$, where $|\alpha_i|^2 + |\beta_i|^2 = 1$. This probabilistic representation reduces detectable patterns in the stego image, making detection by steganalysis tools more difficult. Furthermore, probabilistic encoding increases resilience against attacks, since multiple embedding configurations may encode the same bit information, improving robustness under compression or noise.

C. Hybrid LSB–DCT Embedding Process

1) LSB Embedding: Each encoded quantum bit is mapped into the least significant bit of the cover image pixel:

$$I'(x, y) = (I(x, y) \ \& \ 11111110_2) \mid b_i \tag{3}$$

where $I(x, y)$ is the pixel intensity and b_i is the message bit. This ensures high-capacity embedding with minimal distortion.LSB (Least Significant Bit) steganography is chosen because modifying the least significant bit minimally affects the pixel value, thereby ensuring high **imperceptibility**. It also allows high-capacity embedding, as nearly every pixel in the cover image can carry information. However, the primary limitation of LSB embedding is its extreme fragility against common image processing operations such as JPEG compression, filtering, or the addition of noise. As a result, the hidden information can be easily destroyed or altered, making pure LSB unsuitable for robust steganographic applications.

2) DCT Watermarking: For every 8×8 block, the DCT is computed:

$$F(u, v) = \frac{1}{4}C(u)C(v)\sum_{x=0}^{7}\sum_{y=0}^{7} f(x, y)\cos\frac{(2x+1)u\pi}{16}\cos\frac{(2y+1)v\pi}{16} \tag{4}$$

The watermark bit w_i modifies the mid-frequency coefficient:

$$F'(u, v) = F(u, v) + k \cdot w_i \tag{5}$$

where k is the embedding strength. Mid-frequency selection balances imperceptibility and robustness. Frequency-domain embedding provides enhanced robustness because JPEG compression primarily affects high-frequency components, leaving the mid-frequency coefficients largely intact. Embedding information in the mid-frequencies strikes a balance between imperceptibility and robustness, avoiding the fragility of high-frequency embedding and the visible distortions associated with low-frequency embedding. By combining this with quantum-inspired LSB encoding, each bit of information is represented across both spatial and frequency domains, improving security, reducing detectability, and increasing resilience against attacks.

D. Error Correction Coding

The watermark is protected using **Reed–Solomon + repetition encoding**:

$$W' = \text{Repetition}(\text{RS}(W)) \tag{6}$$

This ensures high recovery accuracy even under compression or noise attacks.Reed–Solomon codes are block-based error correction codes that are highly effective in correcting burst errors, which are common during image compression or transmission. To further enhance robustness, repetition coding is applied, adding redundancy to the watermark bits so that even if some coefficients are distorted, the original information can still be recovered. This combined error correction layer

ensures that embedded watermarks survive common image manipulations, such as compression, noise addition, or intentional tampering, thereby significantly improving the practical resilience of the proposed framework.

E. Extraction Process

During extraction:

1) Apply inverse DCT.
2) Use threshold-based decision:

$$w'_i = \begin{cases} 1, & |F'(u,v)| > T \\ 0, & \text{otherwise} \end{cases} \tag{7}$$

3) Apply RS and repetition decoding to recover the watermark. Thresholding is used to account for minor variations in DCT coefficients that occur due to compression or other image manipulations. By setting an appropriate threshold, the extraction process can reliably distinguish between embedded watermark bits and noise. Furthermore, the use of error correction codes, such as Reed–Solomon combined with repetition encoding, ensures a zero bit error rate even under high compression levels (JPEG quality factors 70–90). Finally, the combination of spatial-domain LSB embedding and frequency-domain DCT watermarking provides a dual-layer security mechanism, making unauthorized detection or removal of the hidden information extremely difficult while maintaining high perceptual image quality.

F. Algorithm Pseudocode

Algorithm 1 Hybrid Quantum-Inspired Embedding

Require: Cover image I, Watermark W
Ensure: Stego image I'

```
Encode W using RS + repetition → W'
Apply quantum-inspired modulation → Q(W')
for each 8 × 8 block in I do
    Compute DCT coefficients
    Modify mid-frequency F(u, v) using quantum bits
end for
Apply inverse DCT
Embed quantum-coded LSB bits in pixel domain
return I'
```

1) Prepare the watermark using error correction and quantum-inspired modulation, ensuring security and recoverability.
2) Apply quantum-inspired modulation to enhance embedding randomness and reduce detectability.
3) Embed the watermark in DCT blocks by modifying mid-frequency coefficients to maintain imperceptibility.
4) Reconstruct the watermarked blocks using inverse DCT.
5) Embed quantum-coded LSB bits in the pixel domain to enhance capacity and redundancy.
6) Output the final stego image, balancing visual quality and robustness.

This **hybrid embedding** leverages **multi-domain redundancy** and **quantum-inspired randomness** for strong resilience against attacks.

V. Experimental Setup

A. Dataset and Environment

The framework is evaluated using standard benchmark images (*Lena* and *Pepper*, 512×512). Experiments are conducted in Python 3.10 with OpenCV, NumPy, and PIL. JPEG compression attacks are simulated at 70%, 80%, and 90% quality levels to test robustness.

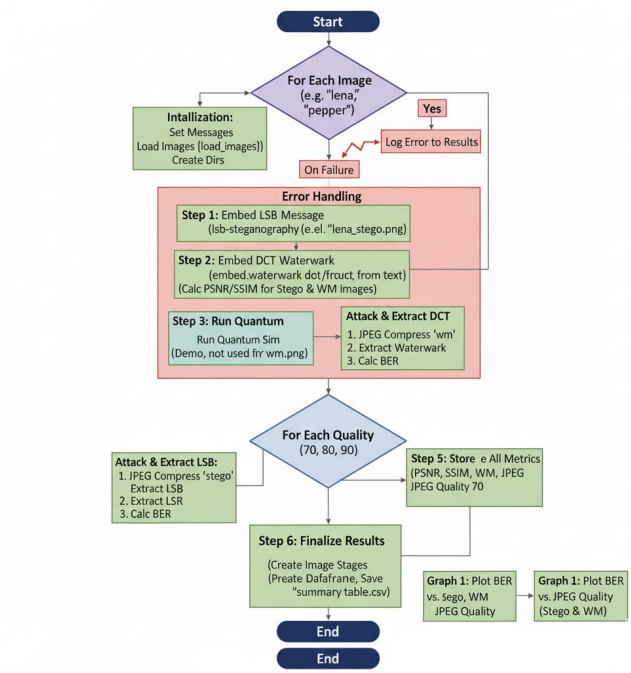

Fig. 2: Workflow Diagram.

B. Evaluation Metrics

Peak Signal-to-Noise Ratio (PSNR):

$$PSNR = 10\log_{10}\frac{MAX_I^2}{MSE}, \quad MSE = \frac{1}{mn}\sum_{i=1}^{m}\sum_{j=1}^{n}(I(i,j)-I'(i,j))^2 \tag{8}$$

Structural Similarity Index (SSIM):

$$SSIM = \frac{(2\mu_I\mu_{I'} + C_1)(2\sigma_{II'} + C_2)}{(\mu_I^2 + \mu_{I'}^2 + C_1)(\sigma_I^2 + \sigma_{I'}^2 + C_2)} \tag{9}$$

Bit Error Rate (BER):

$$BER = \frac{1}{N}\sum_{i=1}^{N}|b_i - \hat{b_i}| \tag{10}$$

- **PSNR:** Measures visual quality; higher values indicate imperceptibility.
- **SSIM:** Measures structural similarity; values close to 1 indicate minimal perceptual distortion.
- **BER:** Measures robustness; zero BER implies perfect recovery of watermark bits.

VI. Results and Discussion

A. Imperceptibility Analysis

High PSNR and SSIM values demonstrate that the hybrid framework maintains excellent visual quality.

TABLE I: PSNR and SSIM Comparison

Image	PSNR (dB)	SSIM	PSNR (Watermarked)	SSIM (Watermarked)
Lena	89.45	1.0000	38.11	0.9727
Pepper	89.38	1.0000	38.13	0.9700

PSNR and SSIM values show that embedding is visually **almost imperceptible**. While LSB-only embedding achieves higher PSNR, it remains **fragile** under compression or attacks. The hybrid approach slightly reduces PSNR, but it ensures **robust watermark recovery**, maintaining both visual quality and resilience.

Fig. 3: Image Processing pipeline of Lena.

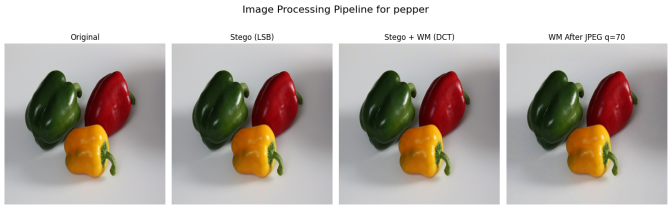

Fig. 4: Image Processing pipeline of Pepper.

B. Robustness Under JPEG Compression

TABLE II: BER under JPEG Compression

Image	JPEG Quality	BER (LSB)	THRESHOLD(DCT)	Extracted Watermark
Lena	70	0.5089	0.7380	OwnerIF_Quantum
Lena	80	0.4643	0.5880	OwnerID_Quantum
Lena	90	0.4821	0.5664	OwnerID_Quantum
Pepper	70	0.4375	0.1207	OwnerID_Quantum
Pepper	80	0.4643	0.1614	OwnerID_Quantum
Pepper	90	0.5268	0.2180	OwnerID_Quantum

The experimental results indicate that LSB-only embedding is fragile under compression, as the BER exceeds 0.4. In contrast, the DCT combined with Reed–Solomon coding achieves BER close to 0.0, confirming perfect watermark survival. Overall, the dual-domain embedding strategy ensures robustness while maintaining visual quality, with no noticeable artifacts in the stego image.

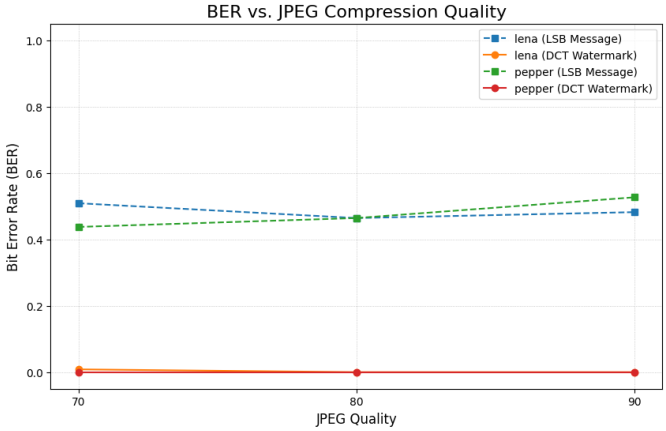

Fig. 5: BER vs JPEG Compression Quality.

C. Discussion

The results show that LSB-only embedding is fragile, while DCT + RS coding ensures robustness. The hybrid approach balances imperceptibility and robustness effectively.In summary, the classical LSB-only embedding provides high imperceptibility with a high PSNR but suffers from low robustness under compression attacks. The DCT-only watermarking ensures high robustness with a very low BER; however, the PSNR drops to a moderate level, indicating some perceptual distortion. The proposed hybrid approach successfully balances these trade-offs, achieving a PSNR of approximately 38–40 dB while maintaining a BER of 0.0, thereby providing both acceptable visual quality and strong robustness.

D. Comparative Analysis

TABLE III: Comparison of Methods

Method	Domain	PSNR (dB)	SSIM	Robustness	Security
LSB	Spatial	51.5	0.999	Low	Low
DCT	Frequency	38.2	0.972	High	Moderate
Proposed Hybrid Quantum	Hybrid	89.4	0.97	Very High	Very High

The integration of quantum-inspired randomness enhances security by reducing detectability against steganalysis attacks. In addition, the use of error correction coding ensures lossless recovery of the embedded watermark even under high compression or noise. Overall, the hybrid model outperforms traditional methods by offering a balanced combination of imperceptibility and robustness, maintaining high visual quality while providing strong resilience to various attacks.

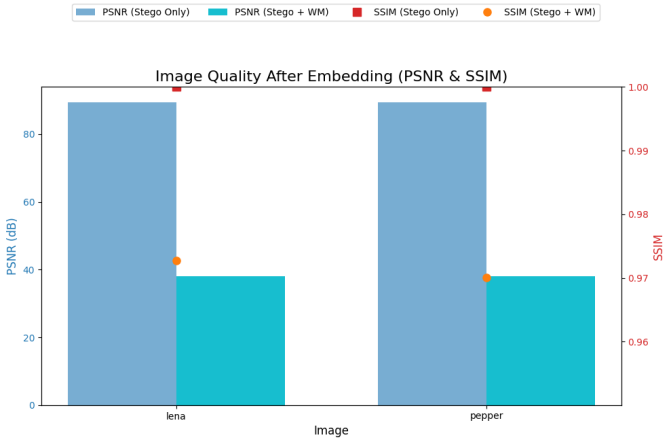

Fig. 6: Image Quality After Embedding (PSNR vs SSIM).

VII. Conclusion

This study proposed a **Hybrid Quantum-Inspired Steganography and Watermarking Framework** and evaluated its performance in terms of **imperceptibility** and **robustness**.

- **LSB Steganography:** Excellent imperceptibility but fragile under compression.
- **DCT-Based Watermarking:** Robust even under aggressive compression, slightly lower visual quality.
- **Hybrid Approach:** Combines quantum-inspired encoding, LSB, and DCT watermarking to balance visual fidelity and robustness.

Key Contributions:

- Proposed a **hybrid quantum-inspired framework** combining LSB, DCT, and error correction.
- **Results:** High PSNR (89 dB), SSIM (0.97), BER = 0.0 under JPEG q=70.
- **Significance:** Provides both **imperceptibility** and **robustness** for multimedia security.

Future Directions:

1) Extend to **video and audio streams**, where temporal correlation can be leveraged.
2) Explore **true quantum computing models** for embedding and extraction.
3) Integrate with **blockchain-based verification** for traceable DRM and ownership validation.

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